

Aditya Saini

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EDUCATION

Purdue University

West Lafayette, IN

B.S. Computer Science (Systems Programming)

Graduation: May 2028 (Expected)

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Mathematics, Computer Architecture, Programming in C, Object-Oriented Programming, Multivariate Calculus

EXPERIENCE

Embedded Engineer

Sep. 2025 – Present

Purdue Electric Racing

West Lafayette, IN

- Developed modular RTOS-based vehicle firmware in C, integrating drivers and performing in-circuit debugging
- Implemented runtime telemetry and data acquisition to validate performance bottlenecks in production firmware
- Designed I2C and SPI HALs for STM32 platforms, standardizing high-frequency sensor communication
- Developed Nxtion LCD UART drivers and optimized command encoding for reliable real-time dashboard updates

Controls Engineer

Dec 2025 – Present

Purdue Space Program

West Lafayette, IN

- Developed a flight dynamics simulation in MATLAB to model rigid body translational and rotational systems
- Implemented and tuned a PID controller for closed-loop stabilization of an unstable nonlinear control system
- Implemented embedded C drivers on ESP32 rocket test platform enabling real-time control and system validation

Robotic Safety Research Assistant

January 2026 – July 2026

PurSec, Purdue Computer Science

West Lafayette, IN

- Developed robotic simulations using AI2THOR to find efficient jailbreaking methods for MLLM controlled robots
- Used AWS EC2 to run and gather results on experiments through EmbodiedBench in a headless environment
- Used Claude Code and Cursor to generate real world environment JSON files to perform robotic tasks

Advanced Mathematics Teacher

June 2023 – July 2025

Kumon North America

Edison, NJ

- Developed tailored pedagogical strategies to explain complex mathematical concepts to a diverse student body, ensuring mastery of core principles for over 20+ concurrent students.
- Managed high-volume academic data entry and performance tracking, utilizing center-specific software to generate progress reports and identify learning bottlenecks for student development.

PROJECTS

Natural Language Shell | C, Python, OpenAI API, Git

June 2025 – January 2026

- Created a lightweight command-line shell in C supporting natural language commands with tokenizer, intent parsing, and bash mapping; integrated OpenAI API for complex commands/queries through OpenRouter.

FRC Team 2554 | Java, OpenCV, Jetson Orin Nano, CUDA

September 2021 – August 2025

- Led the team to its best-ever season as Build President, bringing us to District Championships after 18+ years.
- Developed and led our team's first instance of Swerve Drive, tuned PID/FF loops, and developed auto commands to increase accuracy by 66% using custom vision pipelines to detect AprilTags for robot positioning.
- Managed finances and administrative requirements for the team in conjunction with robot design requirements.

Tava | React Native, Expo, TypeScript, Firebase, Mapbox, OpenAI

September 2025

- Built a cross-platform social app connecting student builders via an AI-powered search and Tinder-style swipe matching, with a node-edge network graph and interactive Mapbox map backed by Firebase auth and data.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, SQL, JavaScript, HTML/CSS, R

Embedded: RTOS, STM32, ESP32, SPI, I2C, UART, CAN, TCP/IP, Arduino, Linux

Developer Tools: Git, Docker, Amazon Web Services, Claude Code, Cursor, Agent Orchestration

Libraries: pandas, NumPy, Matplotlib, React, Node.js, PyTorch, CUDA, OpenCV